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Comparative effectiveness of r-hFSH:r-hLH in a 2:1 ratio versus HP-hMG alone in ovarian stimulation for medically assisted reproduction treatment using IVF/ICSI: A systematic review and meta-analysis

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Abstract

Introduction

Investigation of the number of oocytes recovered from different ovarian stimulation (OS) protocols is essential, as oocyte number is positively associated with the number of good-quality embryos obtained; however, there are few data comparing the number of oocytes retrieved following OS with recombinant or urinary gonadotropins.

Methods

A systematic review and meta-analysis of published evidence was performed to compare the number of oocytes retrieved with recombinant human follicle stimulating hormone:recombinant human luteinizing hormone (r-hFSH:r-hLH) in a 2:1 ratio versus highly purified human menopausal gonadotropin (HP-hMG) alone from Day 1 of OS. The literature search was conducted according to PRISMA guidelines (https://www.prisma-statement.org/) using Medline, EMBASE, Biosis, Scisearch, and TOXCENTER on STN. Studies were included if they reported on r-hFSH:r-hLH in a 2:1 ratio versus HP-hMG alone, both administered from Day 1 of OS, and the results for the number of oocytes retrieved were presented for both interventions. Data were extracted from the full articles independently by two reviewers (JES and BH). Results were analyzed using a random-effects meta-analysis model. Any potential risk of bias was assessed using the ROBINS-I tool. Secondary outcomes were the number of mature (MII) oocytes, clinical pregnancy rate, ongoing pregnancy rate, and live birth rate.

Results

Six studies were included in the meta-analysis (5287 cycles): four non-interventional studies, one non-interventional cross-over study and one randomized controlled trial. The number of oocytes per cycle was significantly higher in women treated with r-hFSH:r-hLH 2:1 compared with those treated with HP-hMG alone (mean difference 1.43, 95% CI 0.18–2.69; P = 0.025). Clinical pregnancy rate per started cycle was significantly higher with r-hFSH:r-hLH 2:1 compared with HP-hMG (rate ratio 1.18, 95% CI 1.06 to 1.33; P = 0.004; five studies). There were insufficient data to draw any firm conclusions on the other secondary outcomes.

Conclusions

The higher number of oocytes retrieved with r-hFSH:r-hLH is a pertinent finding, owing to the well-established positive association between oocyte number and reproductive outcomes. Although we report a higher clinical pregnancy rate for r-hFSH:r-hLH, limited data prevented us from drawing firm conclusions regarding live birth outcomes.

Key words

r-hFSH; r-hLH; HP-hMG; oocyte number; clinical pregnancy rate; systematic review; meta-analysis.

Clinical trial number: not applicable

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Introduction

Exogenous gonadotropins are used during ovarian stimulation (OS) to obtain a supraphysiological number of oocytes. During OS, residual circulating luteinizing hormone (LH) is adequate to support steroidogenesis in the follicles in many cases, thus exogenous follicle stimulating hormone (FSH) alone is considered sufficient for OS (1, 2). However, in the case of patients with LH and FSH deficiency due to reduced gonadotropin production or action (3), residual circulating LH is not sufficient, and exogenous LH in addition to FSH is required for optimal outcomes (3).

Different preparations containing FSH and LH or LH-like activity, including recombinant and urinary-derived products, are used in OS. Originator recombinant human FSH (r-hFSH) and recombinant human LH (r-hLH) are produced using recombinant cDNA technology (4), and are available as separate products (originator r-hFSH; Gonal-f®, Merck Europe BV, Amsterdam, Netherlands; r-hLH; Luveris®, Merck Europe BV, Amsterdam, Netherlands (5, 6)) or as a 2:1 fixed-ratio combination product (r-hFSH:r-hLH 2:1; Pergoveris®, Merck Europe BV, Amsterdam, Netherlands) (7). Based on the cumulative number of treatment cycles between June 2007 (when marketing authorisation was granted) and the end of September 2024 (2,530,049) and using a conservative estimate of the live birth rate (20.3% (2)), an estimated 513,599 babies have been born following co-treatment with r-hFSH:r-hLH 2:1.

The highly purified human menopausal gonadotropin HP-hMG (Menopur®, Ferring Pharmaceuticals, West Drayton, UK) (8)) is obtained from the urine of postmenopausal women and comprises urinary FSH and LH-like activity, which is mainly derived from placental human chorionic gonadotropin (hCG) purified from pregnant women (9).

Identifying the optimum OS protocol is essential for enhancing patient outcomes, as oocyte number is positively associated with the number of good-quality embryos obtained (10), the number of euploid blastocysts (11, 12), the likelihood of pregnancy (13, 14), live birth rates (15-21) and cumulative live birth rates (22-27).

There is published evidence that the combination of FSH and LH may be beneficial in several populations of women, especially those at risk of severe LH and FSH deficiency, which includes women of advanced reproductive age (ARA). A meta-analysis of 12 randomized controlled trials reported that women aged 35–40 years treated with r-hFSH:r-hLH had higher implantation and clinical pregnancy rates versus r-hFSH alone, suggesting that r-hFSH + r-hLH treatment may improve these outcomes in women of ARA (28). In a non-interventional study based on real-world data derived from the Deutsches IVF-Register (D·I·R), treatment with r-hFSH:r-hLH resulted in higher clinical pregnancy and live birth rates compared with r-hFSH alone in women of ARA and in women of ARA with predicted normal ovarian reserve (5–14 oocytes retrieved) (2). Other populations that may benefit from co-administration of r-hFSH and r-hLH compared with r-hFSH alone include women with hypogonadotropic hypogonadism due to either reduced gonadotropin levels or reduced bioactivity, or women with profoundly suppressed LH levels due to GnRH agonist or antagonist treatment during ART, in terms of oocyte quality and implantation rate (29-32), and women who are hyporesponsive to FSH alone (33, 34). Limited data suggest that polymorphisms in *LHB*/*LHCGR* can influence ovarian stimulation outcomes (35, 36). However, while accumulating data suggest that ovarian response to ovarian stimulation may be mediated by various polymorphisms, further studies are needed to investigate the predictive value of these as markers of response in subgroups of women who may require supplementation with LH during ovarian stimulation (36, 37).

In the general population, OS with r-hFSH:r-hLH 2:1 has been reported to yield a higher number of oocytes versus HP-hMG (38-41). Furthermore, in egg donor cycles (42), more oocytes were retrieved after treatment with r-hFSH:r-hLH 2:1 compared with HP-hMG (16.5 vs. 11.8; P = 0.049). In isolation, these few studies do not constitute sufficient evidence for the benefit of r-hFSH:r-hLH 2:1 treatment over HP-hMG with respect to number of oocytes or other reproductive outcomes.

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A previous systematic review of studies found no statistically significant differences between r-hFSH plus r-hLH compared with hMG in clinical pregnancy or live birth rates (43). However, the studies included in this review (43) reported on a heterogenous gonadotropin preparation combinations administered at variable r-hFSH:r-hLH ratios, which makes it difficult to draw firm conclusions from this analysis. Given the preparations currently available, most OS cycles will use r-hFSH and r-hLH in a 2:1 ratio. Therefore, to provide a more precise answer, we conducted a systematic literature review and meta-analysis to assess the difference in the mean number of oocytes retrieved after ovarian stimulation with either rhFSH/rhLH in a fixed 2:1 ratio or HP-HMG.

Methods

Study design

This systematic review and meta-analysis (PROSPERO: CRD42024549520) considered articles published in English, German, Italian, or Spanish between January 2000 and March 2024, regardless of the study design. Studies comparing OS with either r-hFSH:r-hLH in a 2:1 ratio or HP-hMG alone, both administered from Day 1, and reporting the number of oocytes retrieved were included.

Literature search

We conducted the literature search according to PRISMA guidelines (https://www.prisma-statement.org/) using Medline (NLM MEDLINE file), EMBASE (Excerpta Medica), Biosis (BIOSIS Previews/RN Database), SciSearch (Science Citation Index), and TOXCENTER (Toxicology Center Database) on STN. Duplicates were eliminated using an automated procedure. The search was performed as broadly as possible, covering various synonyms for the search terms (**Supplementary Table 1)**. In Medline and EMBASE, the controlled terms were included. To optimize the results, the free-text terms were also searched in abstracts and titles with proximity operators.

Data extraction

The titles and abstract of the studies retrieved were screened for eligibility by JES and BH; if there was disagreement about eligibility, a third reviewer adjudicated. Data on study information (reference, design, demographics/baseline characteristics), treatment information (down-regulation protocol, stimulation dosing, days of stimulation, trigger used, insemination method), and outcome information (number of patients, oocytes recovered for each treatment group, MII oocytes, embryo transfers, implantation rate, clinical pregnancy rate, ongoing pregnancy rate, and live birth) were extracted from the full-text versions of eligible articles, independently, by BH and JES.

Risk of bias assessment

The potential risk of bias from observational studies was assessed using the ROBINS-I tool ("Risk Of Bias In Non-randomized Studies - of Interventions") (44) and visualized using the Risk-of-bias VISualization (robvis) program (45).

Outcomes

The primary outcome was the number of oocytes recovered per cycle. Secondary outcomes were the number of mature (MII) oocytes per cycle, and clinical pregnancy rate, ongoing pregnancy rate, and live birth rate per cycle.

Statistical analyses

The mean (standard deviation) number of oocytes recovered and number of MII oocytes per cycle were analysed using a random-effects model, and are reported as mean difference (95% confidence interval [CI]) for r-hFSH:r-hLH 2:1 versus HP-hMG alone. Clinical pregnancy, ongoing pregnancy , and live birth rates per cycle were also analyzed using a random-effects model and are reported as rate ratios (RR) and 95% CI for r-hFSH:r-hLH 2:1 versus HP-hMG.

To assess variance, the results for the primary and secondary outcomes were weighted using the inverse-variance (IV) method; heterogeneity was estimated using the I^2 statistic and tested using Cochrane's Q chi-square test.

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Sensitivity analyses were performed to assess the impact of excluding studies with high risk of bias. The extent to which each study affected the overall estimate was assessed using the leave-one-out analysis. Additional sensitivity analyses were performed by excluding the studies that used gonadotropin releasing hormone (GnRH) antagonist (39, 41) and by excluding studies in women with diminished ovarian reserve (46), respectively. Publication bias was assessed with Egger's linear regression test to evaluate asymmetry in funnel plots for the outcomes of number of oocytes and clinical pregnancy. All statistical analyses were performed using R (The R Foundation [http://www.r-project.org]).

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Results

Literature review

Eighteen potential studies were identified (shown in Fig. 1). Seven of these studies compared r-hFSH:r-hLH with HP-hMG alone (38-41, 46-48). One study (48) was excluded, as it could not be confirmed that r-hFSH:r-hLH was used in the 2:1 ratio, leaving six studies eligible: four non-interventional (38, 41, 46, 47), one non-interventional cross-over study (39) and one randomized controlled trial (RCT) (40) (**Table 1**). The five non-interventional studies included analysis of previously collected data. The one RCT (Pacchiarotti) included prospective data collection and study-assigned intervention. Details of the OS protocols for each of the studies are reported in **Table 1**.

Primary endpoint

The number of oocytes per cycle was significantly higher in women treated with r-hFSH:r-hLH 2:1 compared with those treated with HP-hMG alone (mean difference 1.43, 95% CI 0.18–2.69; P = 0.025; I^2=98%) (shown in Fig. 2).

No study was considered to have critical bias (i.e., too problematic to provide useful evidence). Four studies (38, 39, 41, 47) were considered to have Low or Moderate bias in the seven domains assessed (shown in Fig. 3). Two studies (40, 46) were judged to have serious bias. Specifically, in Pacchiarotti et al, the reported results were different from the prespecified analysis (Bias domain 7) (40) and in Renzini et al., there was a lack of adjustment for confounding variables (Bias domain 1) (46).

Sensitivity analysis (primary endpoint)

The sequential removal of one study (leave-one-out) produced results consistent with the main analysis (shown in Supplementary Fig. 1). Similarly, when the studies (39, 41) using a GnRH antagonist protocol for pituitary downregulation or the study in women with only diminished ovarian reserve (46) were excluded, respectively, the results were consistent with the main analysis (shown in Supplementary Fig. 2, 3). No reduction in heterogeneity was seen in any of the sensitivity analyses (range 91–99%). The Egger's test of the intercept showed no evidence of publication bias (T = 1.22, P = 0.288).

Secondary endpoints

<u>Number of MII oocytes per started cycle</u>

The number of MII oocytes retrieved did not differ significantly between the two treatment cohorts, with a mean difference of 0.72 (95% CI -0.32 to 1.76; P = 0.177). The I^2 statistic was 57%, indicating moderate heterogeneity among the studies (shown in Fig. 4). However, this outcome was only reported in three studies (38, 39, 41); one study did not report this outcome (47) and two studies only reported the rate and it was not possible to estimate the mean and standard deviation (40, 46).

<u>Clinical pregnancy rate per started cycle after fresh transfer</u>

Clinical pregnancy rates were reported in five studies (38-40, 46, 47). In two studies (39, 40), clinical pregnancy rates were not reported 'per started cycle; however, we were able to calculate the rate per started cycle using data on the number of cycles reported in the articles. One article (Revelli et al (41)) was not included in this outcome, as the clinical pregnancy rate per started cycle was not reported and insufficient data were provided to calculate this outcome. The Egger's test of the intercept showed no evidence of publication bias (T = -1.04, P = 0.374).

No standard definition of clinical pregnancy was cited across studies. No definition was specified in three studies (39, 40, 47), and different definitions of clinical pregnancy were used in two studies . In Fábregues et al, clinical pregnancy was defined by increasing serum concentrations of hCG after embryo transfer and the subsequent demonstration of an intrauterine gestational sac by ultrasonography (38); in Renzini et al, it was defined as ultrasonographic observation of a gestational sac, with or without foetal heartbeat, 25–30 days following embryos transfer (46). Owing to the absence of a standard definition of clinical pregnancy, we deferred to the definitions used in the contributing articles.

Clinical pregnancy rate per started cycle was significantly higher with r-hFSH:r-hLH 2:1 compared with HP-hMG (RR 1.18, 95% CI 1.06 to 1.33; P = 0.004; I^2=10%) (shown in Fig. 4).

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Ongoing pregnancy rate per started cycle following fresh embryo transfer

Ongoing pregnancy rate was reported in only one study (46). No specific definition was included, and the results for ongoing pregnancy rate were not presented directly; therefore, we calculated the ongoing pregnancy rate using data from the reported number of clinical pregnancies minus the reported number of miscarriages. The calculated ongoing pregnancy rate was significantly higher with r-hFSH:r-hLH 2:1 compared with HP-hMG (RR 1.69, 95% CI 1.06 to 2.71; P = 0.028; I^2=N/A) (shown in Supplementary Fig. 4).

Live birth rate per started cycle following fresh embryo transfer

Only two studies reported the live birth rate per started cycle (38, 46), although no definition was specified in either. Live birth rate was not significantly different between r-hFSH:r-hLH 2:1 and HP-hMG (RR 1.40, 95% CI 0.90 to 2.17; P = 0.137; I^2=29%) (shown in Supplementary Fig. 4).

Discussion

In the analysis presented here, OS with r-hFSH:r-hLH in a 2:1 ratio resulted in a significantly higher number of oocytes retrieved compared with HP-hMG. This finding is important, as both the quantity and quality of oocytes influence the outcomes of ovarian stimulation. Furthermore, there is a positive correlation between the number of oocytes retrieved and the number of top-quality embryos, as well as the number of euploid embryos available for fresh transfer or cryopreservation. (10, 49-58).

The clinical pregnancy rate reported was significantly higher after r-hFSH:r-hLH 2:1 treatment in a meta-analysis of five studies, consistent with several studies in specific patient populations. In women with low ovarian reserve (46), the clinical pregnancy per started cycle was higher in the r-hFSH:r-hLH 2:1 group than in the HP-hMG group (12.5 vs. 8.1%, P < 0.02). And using data from the German DIR database in an observational matched case–control study, treatment with r-hFSH:r-hLH 2:1 compared with urinary menopausal gonadotropin with hCG-derived LH-like activity (u-hMG) resulted in higher pregnancy rates per cycle (P = 0.006) and per embryo transfer (P = 0.025), and implantation rate per embryo transferred (P < 0.001) in the r-hFSH:r-hLH 2:1 cohort (47). We were unable to draw any firm conclusions for live birth outcomes owing to the lack of data.

When comparing the number of oocytes retrieved, it is important to consider inherent differences in the formulations of the respective gonadotropins used for OS. Firstly, the recombinant products have a higher level of purity and batch-to-batch consistency than urinary products (59-62) and also have a consistent glycoform profile, which, in the case of FSH, may influence bioactivity (or potency), pharmacokinetic/pharmacodynamic profiles, and clinical efficacy (62). Secondly, HP-hMG also exhibits a high degree of oxidized samples, reaching up to 20% for the alpha subunit (9). Oxidized gonadotropin samples exhibit lower biopotency in both *in-vivo* and *in-vitro* assays (63), potentially impacting clinical effectiveness. Thirdly, differences in the mode of action of the respective components should also be considered.

In-vitro studies demonstrate that although LH and hCG both bind to the same LHCGR receptor on antral follicles, they activate distinct intracellular signalling pathways, consistent with their different physiological roles—LH primarily supporting folliculogenesis and hCG predominantly driving steroidogenesis (64). The action of LH is preferentially exerted via phosphorylated extracellular-regulated kinase 1/2 (pERK1/2) and phosphorylated AKT (also known as protein kinase B), resulting in proliferative/anti-apoptotic signals and partial agonism of progesterone production *in vitro*. This may have a positive effect on oocyte maturation, potentiated by r-hFSH, as well as having an anti-apoptotic effect on cumulus cells and promoting the paracrine signaling required for cell expansion and oocyte signaling during folliculogenesis (65, 66). Conversely, hCG displays notable cAMP/protein kinase A (PKA)-mediated steroidogenic and pro-apoptotic potential (64), which may have a detrimental effect on oocyte development by activating the steroidogenic and pro-apoptotic pathways.

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Collectively, the chemical differences between r-hFSH:r-hLH 2:1 treatment and urinary FSH treatment combined with urinary hCG-derived LH-like activity of placental origin (9), and the differential biological effects outlined above, may explain the higher number of oocytes observed after treatment with rFSH:rLH 2:1 in our study. Our data are in line with results from other studies that consistently showed a higher number of oocytes after OS with r-hFSH, compared with OS with urinary FSH alone (53, 67), after OS with hMG preparations (50, 52, 58, 67-73), or treatment with r-hFSH:r-hLH 2:1 compared with hMG in combination with r-hFSH (74).

Several parameters that have a bearing on OS outcomes could not be accounted for in our analysis, owing to the lack of data available in the source material. Firstly, the starting dose and total gonadotropin dose are an important determinants of the number of oocytes recovered after OS, and there is good evidence that a higher r-hFSH starting dose results in significantly more oocytes recovered in high, normal and low responders compared with a lower r-hFSH starting dose (75). Dose increments of r-hFSH are the only reported significant predictor of the number of oocytes retrieved in the subsequent OS cycle, with an increase of 50 IU over the initial r-hFSH dose leading to an increase of one more oocyte (22), and an increase in the daily gonadotropin dose above 300 IU was also shown to result in a higher mature oocyte yield in women undergoing elective embryo freezing (76). However, we were not able to establish any relationship between dosing and the number of oocytes retrieved (**Supplementary Table 2**), as there was a high degree of variability in dosing among the studies included, and it was unclear whether the reported dosing values represented only the FSH component of the treatment or the activity for the combined components. Secondly, we did not include or exclude studies based on the protocol used to prevent premature LH surge. Only two of the studies used GnRH antagonist protocols (39, 41), which is the current prevailing protocol used for pituitary downregulation (77). We do not envisage this to be of concern, as there is no reported difference in outcomes between the two protocols (78, 79), although we acknowledge this may have implications when generalising the results presented here to the general population and current practice. Thirdly, owing to lack of data in the source publications, we were unable to account for the use of oral contraceptive pill (OCP) pretreatment, which may be employed for several reasons, including the synchronization of follicular development and scheduling of IVF cycles for the convenience of clinicians and patients (80, 81). Nevertheless, current evidence for the effect of pretreatment hormonal contraception on ovarian stimulation is inconsistent (77), and the effect may be dependent on the type of hormonal contraception used (81). Finally, we were unable to account for other factors, such as the use of nutraceutics or other compounds, sperm source or quality, or oocyte developmental competence.

A key strength of this meta-analysis is that the greater number of oocytes retrieved with r-hFSH:r-hLH 2:1 compared with HP-hMG is based on a large number of cycles (2639 for the recombinant and 2648 for the urinary products) using observational and RCT data obtained across four countries. Furthermore, the high degree of similarity in patient baseline characteristics between the two treatment groups within each included study (**Supplementary Table 3**) strengthens the validity of this meta-analysis, and increases the credibility of any differences in outcomes reported.

We also acknowledge several limitations. Firstly, only six studies were deemed eligible for inclusion in the analysis, and only one of these was an RCT. Secondly, the heterogeneity calculation identified that some factors, including patient populations, treatment protocols and dosing patterns, differed among the studies, but important features (i.e., the standard deviations between treatment groups) did not differ within each study. Nevertheless, while the differences across studies warranted sensitivity analyses (including leave-one-out analysis and analyses by downregulation protocol and by ovarian reserve), the results of these were consistent with those for the overall analysis. Thirdly, two studies (40, 46) were judged to have serious reporting risk of bias. In Pacchiarotti et al, the authors planned to correct for the multiple comparisons made on the numerous outcomes by adjusting the significance level that the p values had to reach. However, as the authors presented multiple outcomes as being

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significant without making the p-value adjustment, it is unclear whether the multiple analyses were performed and selected results were presented, rather than all analyses or the prespecified analyses. Since the most significant result (the number of oocytes retrieved) was higher in the r-hFSH:r-hLH 2:1 treatment group, we cannot rule out the possibility that there is bias in the results of this RCT that favors the r-hFSH:r-hLH 2:1 treatment group (40). In Renzini et al, patients in the HP-hMG cohort had a higher baseline antral follicle count (AFC) value compared with the r-hFSH:r-hLH 2:1 cohort (5.9 [2.4] vs. 4.9 [2.1]) but this confounding factor was not adjusted for when the study outcomes were reported, which could have biased the outcome towards a higher number of oocytes in the HP-hMG cohort compared with the r-hFSH:r-hLH 2:1 cohort (46). Fourthly, there was no standard definition of clinical pregnancy rate in the constituent studies, and three studies did not provide any definition; for this endpoint we have deferred to the definitions used by the authors in the source material, and were not able to use the pregnancy definitions according to the ICMART Glossary (82). The use of varying definitions for, and the timing of, clinical pregnancy is a recognized limitation in the literature (83), and we acknowledge that this may have the potential to influence the comparison of pregnancy rates across studies; for example, the use of a broad definition of clinical pregnancy (e.g., the presence of a gestational sac) may lead to higher pregnancy rates compared with studies that adopt a narrower definition (e.g., confirmed fetal heartbeat). However, our analysis is a summary across all studies of the difference in pregnancy rates between treatment groups within the same study. We are confident that a consistent definition of pregnancy was applied to both treatment groups within each study, enabling valid calculation of differences in pregnancy rates within studies and ensuring that the results were not affected by variability in definitions across studies. Additionally, when the study by Pacchiarotti et al (40) was conducted, Italy permitted insemination of three oocytes per IVF cycle only. This may have resulted in a bias towards the selection of good-quality embryos and may have subsequently impacted the clinical pregnancy rate. Finally, quality, stage, and number of transferred embryos were not reported consistently (or were not reported at all), so the potential impact of embryo quality and/or embryo development state on pregnancy rates could not be assessed. Data on several outcomes were limited: only one study reported ongoing pregnancy rate (46) but did not define this outcome or report the result directly, only two studies reported live birth rate (38, 46), and no studies reported on cumulative live birth rate. Egger's test did not reveal evidence of publication bias in the study outcomes; however, the limited number of included studies may have reduced the statistical power to detect such bias.

Conclusions

We identified six studies that utilized r-hFSH:r-hLH in a 2:1 ratio and were included in this meta-analysis. Despite a high level of heterogeneity across the six studies, the results indicated that the number of oocytes retrieved and the clinical pregnancy rate were significantly higher in patients treated with r-hFSH:r-hLH at a 2:1 ratio compared with those treated with HP-hMG. Our analysis has also highlighted both the lack of data for several outcomes (number of MII oocytes, ongoing pregnancy, and live birth rates) as well as inconsistency or absence in the reporting of outcomes such as clinical pregnancy, ongoing pregnancy and live birth. In response to our findings, we call for more high-quality studies, both RCTs and observational studies, to address this gap.

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Declarations

Abbreviations

AFC, antral follicle count
AMH, anti-Müllerian hormone
CI, confidence interval
D, domain
FSH, follicle stimulating hormone
hCG, human chorionic gonadotropin
HP-hMG, highly purified human menopausal gonadotropin
ICSI, intracytoplasmic sperm injection
IU, international units
IVF, in vitro fertilisation
LH, luteinizing hormone
MII, metaphase II
N, number
N/A, not applicable
NIS, non-interventional study
NR, not reported
OS, ovarian stimulation
PGT, preimplantation genetic testing
POR, poor ovarian response
PR, pregnancy rate
RCT, randomised controlled trial
r-hFSH, recombinant human follicle stimulating hormone
r-hLH, recombinant human luteinizing hormone
ROBINS-I, Risk Of Bias In Non-randomized Studies - of Interventions- I
RR, rate ratios
SD, standard deviation
u-hMG, urinary human menopausal gonadotropin

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Ethics approval and consent to participate

Not applicable, as this systematic review and meta-analysis used data from previously published literature.

Consent for publication

Not applicable, as this systematic review and meta-analysis used data from previously published literature.

Conflict of interests

MHD has received fees Merck Healthcare KGaA, Ferring Inc, and Organon Inc for consultations and scientific presentations. MHD is the Editor and Chief of the Journal Clinical and Experimental Obstetrics and Gynecology. MHD has published several textbooks related to infertility with diverse publishers.
BH has received fees from Merck Healthcare KGaA as a statistical consultant**.**
RF has received honoraria for lectures from Merck Healthcare KGaA, Medea and Event Planet.
SCE has received unrestricted research grants from Merck KGaA; consulting fees from Merck, Medea, and Event Planet; payment or honoraria for lecture fees from Merck, Sanitanova, Medea and Event Planet; is an advisory board member for Nature Reviews Urology (unpaid); and is Head, Andrology

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Committee, Brazilian Society of Human Reproduction (unpaid), and Topic Leader of the Male Infertility Group, WHO Infertility Guidelines (unpaid).
MC declares no conflicts of interest.
KMS declares no conflicts of interest.
LG has received honoraria for external consultant from CooperSurgical
DS has received honoraria for lectures from Merck Healthcare KGaA, Medea and Event Planet
IF declares no conflicts of interest.
CA declares receipt of personal fees from Merck, Medea, Event Planet, IBSA, Ely Lilly, S&R Farmaceutic, and Biogen; and payment or honoraria from Merck, Medea, Event Planet, IBSA and Ferring.
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Author Contributions
MHD: Interpretation of the data; drafting the article and critically revising it for important intellectual content; final approval for the article to published; accountability for all aspects of the work.
JES: Conception and design of the study; data acquisition; drafting the article and critically revising it for important intellectual content; final approval for the article to published; accountability for all aspects of the work.
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Data availability

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Any requests for data by qualified scientific and medical researchers for legitimate research purposes will be subject to Merck KGaA's Data Sharing Policy. All requests should be submitted in writing to Merck KGaA's data sharing portal (https://www.merckgroup.com/en/research/our-approach-to-research-and-development/healthcare/clinical-trials/commitment-responsible-data-sharing.html). When Merck KGaA has a co-research, co-development, or co-marketing or co-promotion agreement, or when the product has been out-licensed, the responsibility for disclosure might be dependent on the agreement between parties. Under these circumstances, Merck KGaA will endeavor to gain agreement to share data in response to requests.

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Figure Legends

Fig. 1. PRISMA diagram of the assessment of studies to be included in the analysis

Abbreviations: hMG, human menopausal gonadotropin; IVF, in-vitro fertilization; r-hFSH, recombinant human follicle-stimulating hormone; r-hLH, recombinant human luteinizing hormone.

Fig. 2. Forest plot of the number of oocytes retrieved per cycle

Abbreviations: CI, confidence interval; hMG, human menopausal gonadotropin; r-hFSH, recombinant human follicle-stimulating hormone; r-hLH, recombinant human luteinizing hormone; SD, standard deviation.

Fig. 3. Risk of bias assessment for number of oocytes retrieved per cycle

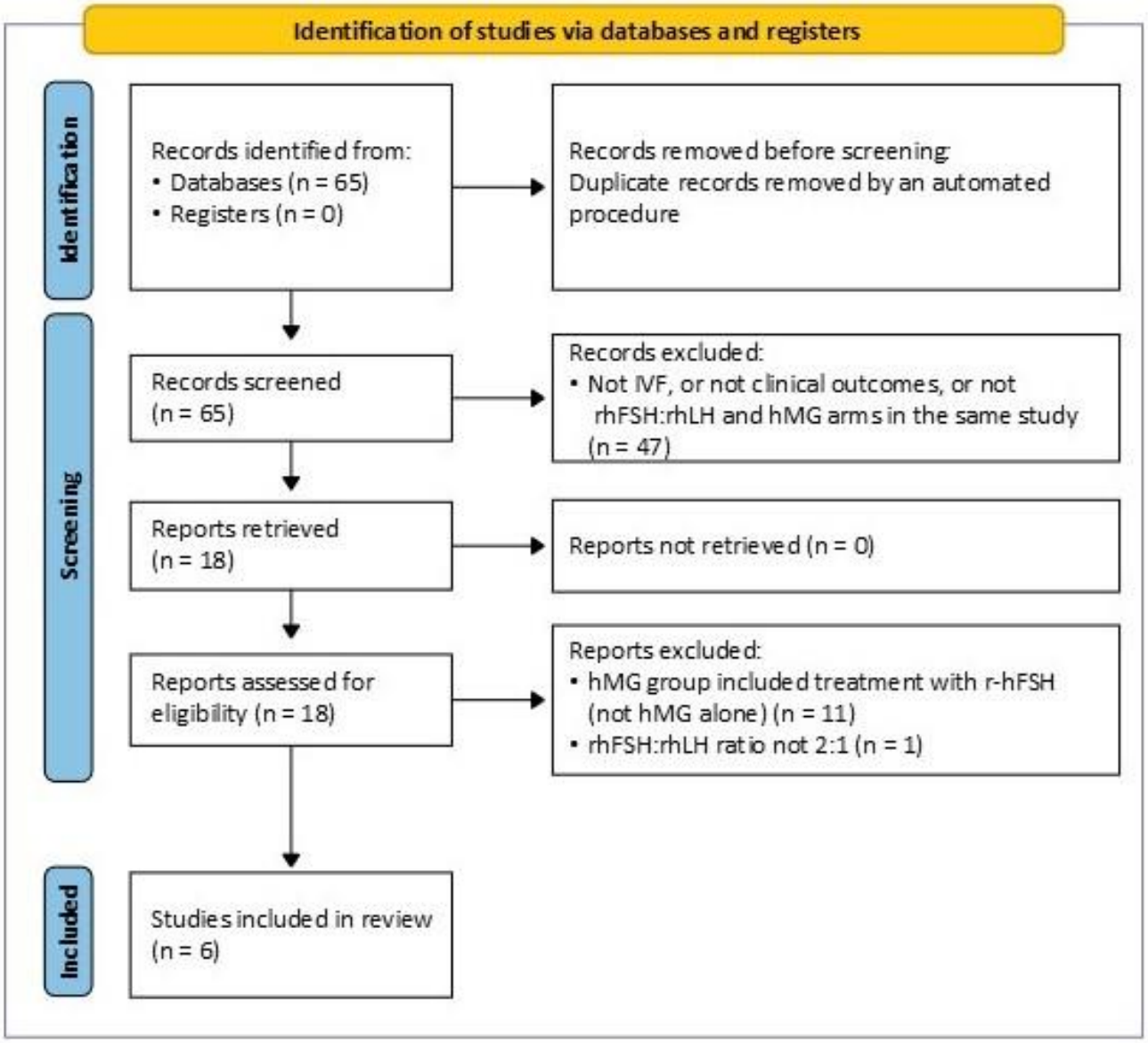
Abbreviations: D, domain; D1, bias due to confounding; D2, bias due to the selection of participants; D3, bias in the classification of interventions; D4, bias due to deviations from the intended interventions; D5, bias due to missing data; D6, bias in measurement of outcomes; D7, bias in the selection of the reported result.

Fig. 4. Forest plots of secondary outcomes: MII oocytes (A), clinical pregnancy rate (B)

Abbreviations: CI, confidence interval; CPR, clinical pregnancy rate; hMG, human menopausal gonadotropin; MII, metaphase II oocytes; r-hFSH, recombinant human follicle-stimulating hormone; r-hLH, recombinant human luteinizing hormone; SD, standard deviation.

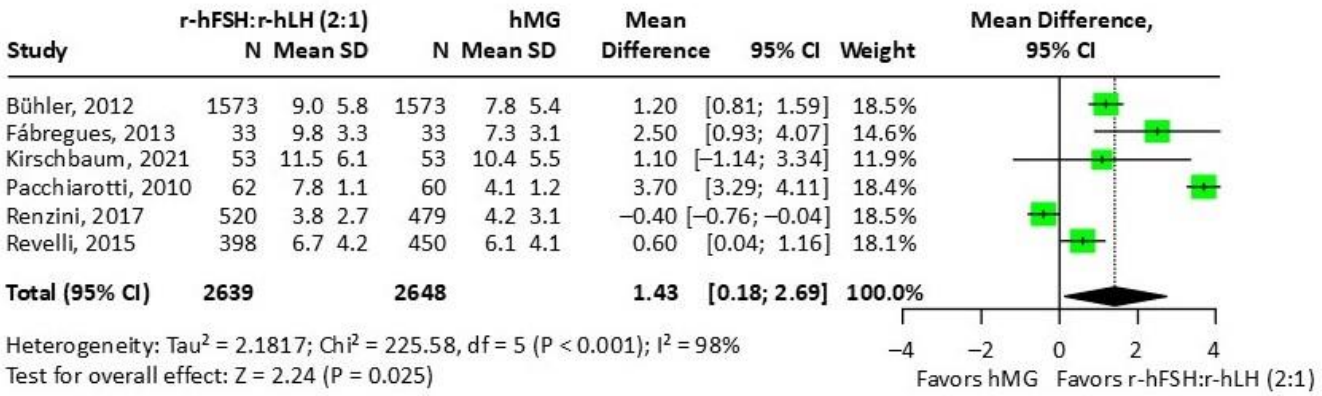
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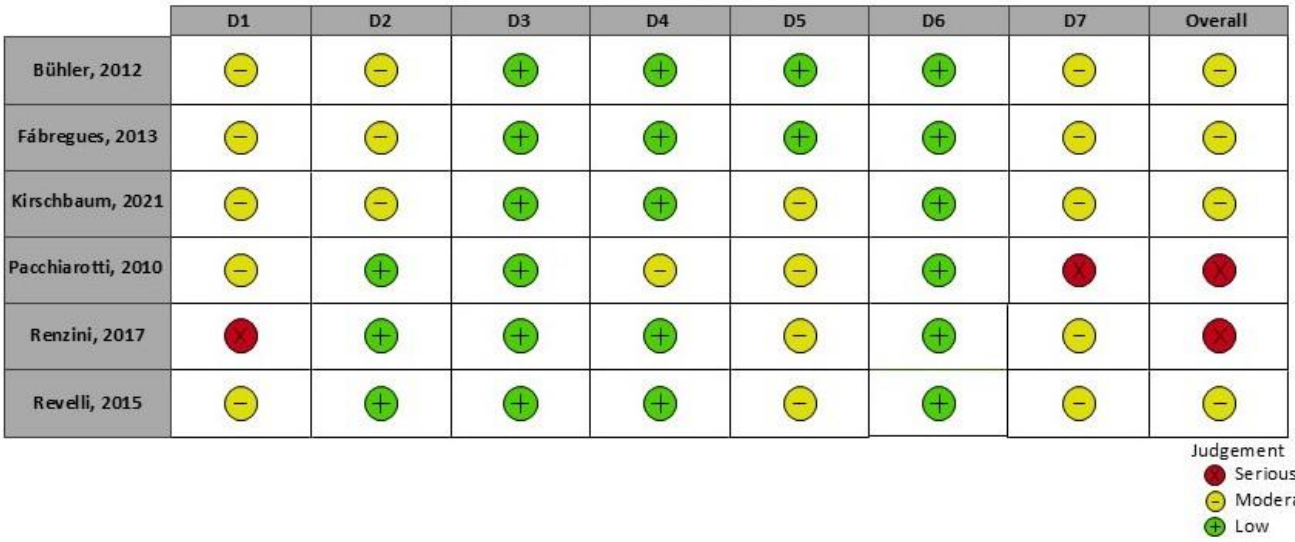


Heterogeneity: $Tau^2 = 2.1817$; $Chi^2 = 225.58$, df = 5 ($P < 0.001$); $I^2 = 98\%$
Test for overall effect: Z = 2.24 (P = 0.025)

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Risk of bias domains



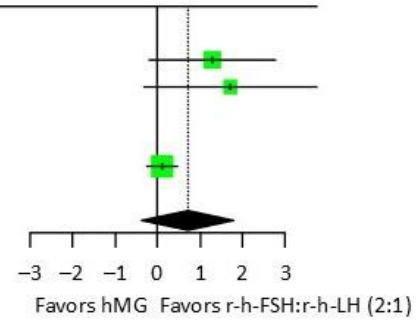
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A

Study	r-hFSH:r-hLH (2:1) N	Mean	SD	hMG N	Mean	SD	Mean Difference	95% CI	Weight
Bühler, 2012	1573	.	.	1573	.	.			0.0%
Fábregues, 2013	33	7.3	2.8	33	6.0	3.2	1.28	[–0.20; 2.76]	27.2%
Kirschbaum, 2021	53	10.0	5.8	53	8.3	4.6	1.70	[–0.32; 3.72]	18.5%
Pacchiarotti, 2010	62	.	.	60	.	.			0.0%
Renzini, 2017	520	.	.	479	.	.			0.0%
Revelli, 2015	398	4.3	2.7	450	4.2	2.3	0.10	[–0.24; 0.44]	54.2%
Total (95% CI)	**2639**			**2648**			**0.72**	**[–0.32; 1.76]**	**100.0%**

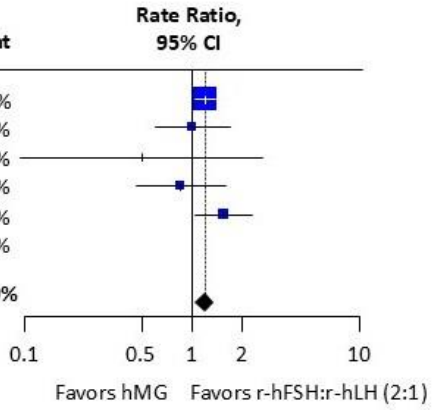
Heterogeneity: $Tau^2 = 0.4911$; $Chi^2 = 4.64$, df = 2 (P = 0.098); $I^2 = 57\%$
Test for overall effect: Z = 1.35 (P = 0.177)



B

Study	r-hFSH:r-hLH (2:1) N	CPR	hMG N	CPR	Rate Ratio	95% CI	Weight
Bühler, 2012	1573	25.5%	1573	21.5%	1.19	[1.05; 1.35]	81.4%
Fábregues, 2013	33	48.5%	33	48.5%	1.00	[0.61; 1.64]	5.3%
Kirschbaum, 2021	53	3.8%	53	7.5%	0.50	[0.10; 2.61]	0.5%
Pacchiarotti, 2010	62	24.2%	60	28.3%	0.85	[0.47; 1.55]	3.7%
Renzini, 2017	520	12.5%	479	8.1%	1.54	[1.05; 2.24]	9.2%
Revelli, 2015	398	NA%	450	NA%			0.0%
Total (95% CI)	**2639**		**2648**		**1.18**	**[1.06; 1.33]**	**100.0%**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 4.46$, df = 4 (P = 0.347); $I^2 = 10\%$
Test for overall effect: Z = 2.90 (P = 0.004)



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Table 1: Summary of studies included in the systematic review and meta-analysis

Reference	Country	Design	Ovarian reserve	Ages	Total number of cycles started r-hFSH:r-hLH 2:1	Total number of cycles started HP-hMG	GnRH protocol	Oocytes retrieved data available	Oocyte maturation trigger	MAR	Fresh/frozen transfer	Day of embryo transfer
Buhler, 2012 (37)	Germany	NIS	All	All and subgroups <35 and ≥35 years)	1573	1573	Agonist	Yes	Not reported	IVF/ICSI	Not reported	Not reported
Fábregues, 2013 (28)	Spain	NIS	Normal	All	33	33	Agonist	Yes	hCG	ICSI	Fresh	Day 2–3
Kirshenbaum, 2021 (29)	Israel	NIS - crossover	All	All	53	53	Antagonist	Yes	hCG/agonist/ both	IVF/ICSI	Fresh	Day 2-3
Pacchiarotti, 2010 (30)	Italy	RCT	Normal	All	62	60	Agonist	Yes	hCG	ICSI	Fresh	Day 2
Renzini, 2017 (36)	Italy	NIS	Diminished (AFC < 11 and AMH <= 1.1 ng/ml)	≤42	520	479	Agonist	Yes	hCG	ICSI	Fresh	Day 2–3
Revelli, 2015 (31)	Italy	NIS	All	All	398	450	Both	Yes	hCG	IVF/ICSI	Fresh	Not reported

AFC, antral follicle count; AMH, anti-Müllerian hormone; GnRH, gonadotropin-releasing hormone; hCG, human chorionic gonadotropin; HP-hMG, highly purified menopausal gonadotropin; ICSI, intracytoplasmic sperm injection; IVF, in-vitro fertilization; MAR, medically assisted reproduction; NIS, non-interventional study; RCT, randomized controlled trial; r-hFSH, recombinant human follicle-stimulating hormone; r-hLH, recombinant human luteinizing hormone.

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